



***B.Tech Computer
Science and Engineering
(Cyber Security & Forensics)***

Division	Faculty of Engineering and Technology
School Name	School of Computer Science & Engineering
Department Name	Department of Computer Engineering and Technology
Programme Name	B.Tech Computer Science and Engineering (Cyber Security & Forensics)

Semester	Odd (I)	Even (II)	Total Credits
First Year	20	21	41
Second Year	22	22	44
Third Year	22	22	44
Fourth Year	22	14	36

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| Course Type | Description |
|----------------------|---|
| Programme Core | Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU |
| Programme Electives | Open electives under the programme allow students to specialise in a particular area connected to their major.. |
| University Core | Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students. |
| University Electives | Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core. |

Semester I

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| I | UC | Indian Constitution | 1 |
| I | UC | Environment and Sustainability | 1 |
| I | UC | Yoga - I | 1 |
| I | UC | Social Leadership Development Program | 1 |
| I | UC | Financial Literacy | 1 |
| I | UC | Python Programming | 2 |
| I | PF | Linear Algebra and Differential Calculus | 3 |
| I | PF | Engineering Chemistry | 3 |
| I | PF | Engineering Physics | 3 |
| I | PF | Ideas and Innovations in Manufacturing | 1 |
| I | PF | Engineering Graphics | 3 |
| | | Total Credits: | 20 |

Semester II

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| II | UC | Yoga - II | 1 |
| II | UC | Co-creation | 1 |
| II | UC | Foundation of Peace | 2 |
| II | UC | Indian Knowledge System (General) | 2 |
| II | UC | Sports | 1 |
| II | PF | Engineering Mechanics | 3 |
| II | PF | Foundations of Programming | 4 |
| II | PF | Discrete Mathematics with Graph Theory | 3 |
| II | Program Capstone Project/
Seminar and Internships | Foundations of Computer Architecture
and System Design | 4 |
| | | Total Credits: | 21 |
| | | | |

Semester - III

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| III | UC | Spiritual and Cultural heritage: Indian Experience | 2 |
| III | UC | Research Innovation Design Entrepreneurship (RIDE) | 1 |
| III | UE | University Electives - I | 3 |
| III | PF | Calculus and Numerical Methods | 4 |
| III | PM | Object Oriented Programming using C++ and Java | 1 |
| III | Program Capstone Project/
Seminar and Internships | Object Oriented Programming Laboratory | 1 |
| III | PF | Data Structures | 3 |
| III | Program Capstone Project/
Seminar and Internships | Data Structures Laboratory | 1 |
| III | PM | Database Management System | 3 |
| III | PM | Database Management System Laboratory | 1 |
| III | PM | Software Engineering and Project Management | 2 |
| | | Total Credits | 22 |

SEMESTER - IV

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| IV | PM | Computer Networks | 3 |
| IV | PM | Computer Networks Laboratory | 1 |
| IV | UC | Rural Immersion | 1 |
| IV | UC | Life Transformation Skills | 1 |
| IV | UE | University Electives - II | 3 |
| IV | PF | Probability and Statistics | 4 |
| IV | PM | Advanced Data Structures | 3 |
| IV | Program Capstone Project/
Seminar and Internships | Advanced Data Structures Laboratory | 1 |
| IV | Program Capstone Project/
Seminar and Internships | Cognitive Skills | 1 |
| IV | PF | Indian Knowledge System (Sci. & Tech.) | 2 |
| IV | PF | Embedded Systems and Internet of
Things Laboratory | 1 |
| IV | Program Capstone Project/
Seminar and Internships | Seminar | 1 |
| | | Total Credits: | 22 |
| | | | |

Semester - V

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| V | PF | Information and Cyber Security | 4 |
| V | UC | Managing Conflicts Peacefully: Tools and Techniques | 2 |
| V | UE | University Electives - III | 3 |
| V | PE | Program Elective - I | 4 |
| V | PM | Operating System | 3 |
| V | PM | Operating System Laboratory | 1 |
| V | PM | Artificial Intelligence and Machine Learning Techniques | 4 |
| V | Program Capstone Project/Seminar and Internships | Full Stack Development Laboratory | 1 |
| | | Total Credits: | 22 |

Semester - VI

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| VI | PM | Digital Forensics and Investigation | 3 |
| VI | PM | Digital Forensics and Investigation Laboratory | 1 |
| VI | UC | National Academic Immersion Program | 2 |
| VI | PE | Program Elective - II | 4 |
| VI | PM | Design and Analysis of Algorithms | 3 |
| VI | PM | Theory of Computation | 3 |
| VI | PM | Vulnerability Assessment and Penetration Testing | 3 |
| VI | PM | Vulnerability Assessment and Penetration Testing Laboratory | 1 |
| VI | Program Capstone Project/Seminar and Internships | Mini Project | 1 |
| VI | Program Capstone Project/Seminar and Internships | Competitive Coding | 1 |
| | | Total Credits | 22 |

Semester - VII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| VII | PM | Security Management, Cyber Laws and Ethics | 2 |
| VII | PE | Program Elective - III | 4 |
| VII | Program Capstone Project/Seminar and Internships | Capstone Project | 6 |
| VII | PM | Cloud Infrastructure and Security | 2 |
| VII | PM | Cloud Infrastructure and Security Laboratory | 1 |
| VII | PM | Blockchain Technology Concepts and Application | 2 |
| VII | PM | Blockchain Technology Laboratory | 1 |
| VII | PM | System Software and Compiler Design | 4 |
| | | Total Credits: | 22 |

Semester - VIII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|----------------------------|---------------|
| VIII | PE | Program Elective - IV | 4 |
| VIII | Program Capstone Project/Seminar and Internships | Internship | 10 |
| | | Total Credits: | 14 |

Program Electives

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| V | PE - I | Bigdata Technologies | 4 |
| V | PE - I | Data Privacy | 4 |
| V | PE - I | Enterprise Infrastructure Security | 4 |
| V | PE - I | Identity and Access Management | 4 |
| VI | PE - II | Augmented Reality and Virtual Reality | 4 |
| VI | PE - II | Wireless and Mobile Device Security | 4 |
| VI | PE - II | Incident Response and Malware Analysis | 4 |
| VI | PE - II | Cyber Threat and Intelligence Management | 4 |
| VII | PE - III | High Performance Computing | 4 |
| VII | PE - III | Cyber Physical System Security | 4 |
| VII | PE - III | Data Science for Cybersecurity and Forensics | 4 |
| VII | PE - III | Risk Assessment and Security Policies | 4 |
| VIII | PE - IV | User Interface and User Experience Design | 4 |
| VIII | PE - IV | Security and Privacy in Social Media | 4 |
| VIII | PE - IV | Application and Web Security | 4 |
| VIII | PE - IV | Attack, Reporting and Documentation | 4 |

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.